

KernelCI kci-dev: Closing the Developer Feedback Loop

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20 minutes

- ~10 minutes: quick kci-dev intro & demo-style overview
- ~10 minutes: roadmap discussion and open Q&A

Who this talk is for

- If you are in this MC you are probably interested in Kernel testing
- Make it easier with **kci-dev**

Motivation: Why the feedback loop isn't closed (yet)

- Massive combination of trees, branches, architectures, and tests
- **KernelCI** provides tons of data, but primarily via dashboards and APIs
- Developers still rely on email reports or ad-hoc scripts to get results
- Friction remains in getting CI feedback directly into dev workflow

KernelCI in one slide

- Upstream, open testing platform for the Linux kernel
- Builds, boots, and tests across distributed labs (continuous integration at scale)
- **Volume:** thousands of kernel builds and boot tests each week
- Interaction historically via web UI or raw JSON APIs – not developer-friendly
- **Need:** an ergonomic way to query and trigger tests from the terminal

What kci-dev lets you do today

- **Query results:** Get build/boot summaries, compare test results between revisions, filter by hardware, etc. – all via one CLI
- **Trigger tests:** Launch KernelCI jobs for your tree or branch on demand and do simple bisection (no waiting for nightly runs)
- **Fetch logs quickly:** Download and inspect boot logs or test output straight from the terminal
- **Integrate easily:** Scriptable JSON outputs for CI pipelines, plus shell completions for interactive use

Real-World Context

1. Developer Tree

- Trigger KernelCI jobs for any tree in KernelCI
- Example: using `kci-dev checkout` with `--giturl`, `--branch`, `--tipoftree`

2. CIP SLTS kernels

- Trigger SLTS kernel builds for CIP trees
- Example:

```
kci-dev checkout --giturl https://git.kernel.org/.../cip/linux-cip.git --branch linux-6.1.y-cip --tipoftree
```
- Validate long-term maintenance CI status directly from the terminal

3. ChromiumOS coverage

- Use `kci-dev maestro coverage` to retrieve kernel coverage info
- Terminal shows function/line coverage and URLs to the reports
- Graph views generated through `--graph-output` and opened locally

4. Gentoo Kernel (gentoo-sources)

- Validate Gentoo patchset kernels using KernelCI
- Run builds/tests against gentoo-sources trees before publication
- Compare revisions using `kci-dev results compare`

Triggering builds with job filter

```
kci-dev checkout \  
  --giturl https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git \  
  --branch master \  
  --tipoftree \  
  --job-filter baseline-arm64-foundriesio \  
  --watch \  
  --test crit  
Retrieving latest commit on tree: https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git branch: master  
Commit to checkout: d358e5254674b70f34c847715ca509e46eb81e6f  
OK  
treeid: 98a7e00aae69091a4923f00f2f42508b4c2558c962b797bafa661ceadc8063b1  
checkout_nodeid: 693af331fee8383e92b6a864  
Watching for jobs on treeid: 98a7e00aae69091a4923f00f2f42508b4c2558c962b797bafa661ceadc8063b1  
Watching for test result: crit  
job_filter: baseline-arm64-foundriesio, checkout  
Running job....  
PASS branch checkout - node_id:693af331fee8383e92b6a864 (2025-12-11T16:39:06.510000)  
Running job.....
```

Remaining gaps in developer workflow

- **Maintainer adoption:** Not all maintainers use kci-dev yet – some workflows (e.g. patch series on mailing lists) aren't covered
- **Context switching:** Still some need to visit dashboards for certain info (e.g. detailed triage or artifact browsing)
- **Incomplete features:** New KernelCI data (like bisect info) need better CLI support or docs
- **Performance concerns:** Large result sets can be slow to fetch; no offline caching means repeated queries are inefficient

Technical debt & under-the-hood improvements

- **Refactor and modularize:** Clean up duplicate logic (e.g. consolidate Maestro API code) to simplify maintenance
- **Improve test coverage:** Add more unit and integration tests for kci-dev itself to catch regressions early
- **Robustness:** Handle API changes gracefully (ensure kci-dev keeps working as KernelCI evolves)
- **Library-ready structure:** Decouple CLI parsing from core logic so kci-dev functions can be imported and reused easily

Future features on the roadmap

- **Ad-hoc patch testing:** Trigger KernelCI runs on patch series or unmerged code (e.g. from an mbox or local branch)
- **Smarter result diffing:** Richer comparisons between test runs (highlight regressions, link to relevant logs, inline dashboard links for context)
- **Local caching of data:** Cache recent results locally to speed up repeated queries and allow basic offline result browsing
- **Deeper Git integration:** Automatically infer context (current git repo/branch) for commands and potentially integrate with git workflows (e.g., run tests on `git push`)

Team priorities & milestones

- **Make kci-dev maintainer and developers friendly:** Ensure kernel tree maintainers can seamlessly use it for their workflows (multiple branches, stable releases, etc.)
- **Enable library usage:** Provide a supported Python API so kci-dev can be imported as a module in other tools or CI systems
- **Stability towards v1.0:** Focus on fixing bugs and polishing UX for a stable 1.0 release (address known issues, consistent outputs)
- **Stay in sync with KernelCI:** Align with KernelCI's new pipeline and features, so kci-dev remains fully compatible as the backend evolves

Questions for Kernel Developers & Maintainers

- What slows you down today?
- Do you need CI results earlier in your workflow? How early?
- What data from KernelCI do you *actually* look at during development?
- Would patch-series testing (mbox / git range) help your workflow?
- How often do you compare test results between revisions?
- Do you want CI to auto-run on your tree, or only on-demand?
Tree registration on KernelCI or local patches? self-service
maestro deploy method?
- What's missing in kci-dev to make you use it daily?



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Getting started after this talk

- **Give kci-dev a try** on your own kernel tree this week (pip install kci-dev and go!)
- **Share your feedback:** Identify your top pain points in kernel testing and let us know if kci-dev doesn't solve them yet
- **Join the community:** Come to KernelCI community meetings or chat – help us prioritize what matters to you
<https://discord.gg/KWbrbWEyqb>
- **Contribute:** Whether it's a bug report, a documentation fix, or a code patch, all contributions are welcome and valued

Thank You !

- kci-dev

Documentation: <https://kci.dev>

GitHub: <https://github.com/kernelci/kci-dev/>

Your feedback will help shape the future of kci-dev

- slides

Slides:

https://aliceinwire.github.io/presentations/2025/LPC_2025_TALK/